

**TECHNICAL
OCCUPATIONAL**

WELDING YEAR 3 TERM 4

NAME OF LEARNER

CHAPTER 1

Safety and First Aid, HIV and AIDS procedures

Safety Procedures:

1. Identify potential hazards in the workplace.

Types of Hazards		
a)	Physical Hazards:	- Slips, trips, and falls - Electrical hazards - Fire hazards - Noise hazards - Vibration hazards
b)	Chemical Hazards:	- Toxic substances - Flammable liquids - Corrosive materials - Carcinogenic substances
c)	Biological Hazards:	- Infectious diseases - Bloodborne pathogens - Biological agents
d)	Ergonomic Hazards:	- Repetitive strain injuries - Musculoskeletal disorders - Poor lighting and ventilation
e)	Psychological Hazards:	- Stress and anxiety - Bullying and harassment - Workload and pressure

Identifying Hazards

	Identifying Hazards	How To Identify Hazards
a.	Conduct Regular Risk Assessments:	- Walk-through inspections - Employee surveys and feedback - Review of incident reports
b.	Look for Potential Hazards:	- Unstable structures or equipment - Unsecured chemicals or materials - Poor housekeeping or maintenance - Inadequate training or supervision
c.	Consider Employee Factors:	- Physical limitations or disabilities - Language barriers or communication issues - Inexperience or lack of training
d.	Review Incident Reports and Near Misses:	- Identify root causes and patterns - Implement corrective actions and preventative measures

CONTROL AND MITIGATION:

1. Eliminate or Substitute Hazards:

- Remove or replace hazardous materials or equipment
- Implement safer alternatives or processes



2. Engineering Controls:

- Install safety guards or barriers
- Implement ventilation systems or noise reduction measures

3. Administrative Controls:

- Develop and enforce safety policies and procedures
- Provide training and supervision

4. Personal Protective Equipment (PPE):

- Provide and ensure proper use of PPE (e.g., gloves, safety glasses)

ACTIVITY 1

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

IDENTIFYING POTENTIAL HAZARDS IN THE WORKPLACE

QUESTION 1

What is the first step in identifying potential hazards in the workplace?

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QUESTION 2

What type of hazard can cause musculoskeletal disorders?

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QUESTION 3

What is the purpose of reviewing incident reports?

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QUESTION 4

Why is it important to consider employee factors when identifying potential hazards?

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QUESTION 5

What is the difference between a hazard and a risk?

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QUESTION 6

How can regular risk assessments help identify potential hazards?

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QUESTION 7

What control measure can be implemented to eliminate a chemical hazard?

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QUESTION 8

How can personal protective equipment (PPE) be used to control hazards?

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QUESTION 9

What administrative control can be implemented to reduce the risk of injury from a hazardous task?

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QUESTION 10

What are the potential consequences of not identifying and controlling hazards in the workplace?

QUESTION 11

How can incident reports be used to identify trends and patterns in workplace hazards?

QUESTION 12

What are the benefits of involving employees in the hazard identification process?

QUESTION 13

How can the workplace be designed to reduce the risk of hazards?

Demonstrate proper use of Personal Protective Equipment (PPE).

	Types of PPE	PPE	Image
1.	Head Protection	Hard hats, safety helmets	
2.	Eye Protection	Safety glasses, goggles, face shields	
3.	Ear Protection	Earplugs, earmuffs	
4.	Hand Protection	Gloves, safety gloves	

5.	Foot Protection	Safety shoes, boots, steel-toed shoes	
6.	Body Protection	Coveralls, aprons, lab coats	
7.	Respiratory Protection	Dust masks, respirators	

Demonstration on how to properly wear PPE.

Step-by-step guide on the proper use of Personal Protective Equipment (PPE) in a welding environment:

PPE COMPONENTS:

1. Safety Glasses/Goggles
2. Welding Helmet
3. Hearing Protection (Earplugs/Earmuffs)
4. Welding Gloves
5. Long-Sleeved Shirt/Pants
6. Steel-Toed Boots
7. Respiratory Protection (Optional)

PROPER USE OF PPE:

STEP 1: PRE-WELDING CHECKLIST

1. Ensure all PPE is in good condition.
2. Check expiration dates for disposable PPE.
3. Verify proper fit and adjustment.

STEP 2: DONNING PPE

1. Put on safety glasses/goggles.
2. Wear welding helmet with shade lens (adjust to correct shade).
3. Insert earplugs or wear earmuffs.
4. Put on welding gloves.
5. Wear long-sleeved shirt/pants and steel-toed boots.

STEP 3: DURING WELDING

1. Keep PPE in place at all times.
2. Adjust welding helmet shade as needed.
3. Monitor hearing protection.

STEP 4: POST-WELDING

1. Remove PPE carefully.
2. Clean and store PPE properly.
3. Inspect PPE for damage.

COMMON PPE MISTAKES TO AVOID:

1. Not wearing PPE at all times during welding.
2. Incorrectly adjusting welding helmet shade.
3. Not replacing expired or damaged PPE.
4. Not following manufacturer guidelines.

CONSEQUENCES OF NOT USING PPE:

1. Eye damage (arc eye, retinal burns).
2. Hearing loss.
3. Skin burns.
4. Respiratory problems.

Best Practices:

Make PPE a habit

Lead by example

Provide regular training and reminders

Encourage employee feedback and suggestions

Continuously monitor and evaluate PPE effectiveness

ACTIVITY 2

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Proper use of Personal Protective Equipment (PPE):

QUESTION 1

What is the primary purpose of wearing PPE?

- a) To protect others from harm
- b) To protect yourself from harm
- c) To comply with regulations
- d) To improve productivity

QUESTION 2

Which type of PPE is used to protect the eyes?

- a) Safety glasses
- b) Earplugs
- c) Gloves
- d) Steel-toed shoes

QUESTION 3

How often should PPE be inspected?

- a) Daily
- b) Weekly
- c) Monthly
- d) Annually

QUESTION 4

What is the correct way to store PPE when not in use?

- a) In a designated area
- b) In a toolbox
- c) On the floor
- d) In a trash can

QUESTION 5

Why is it important to wear PPE consistently?

- a) To avoid getting in trouble
- b) To protect yourself from harm
- c) To improve productivity
- d) To comply with regulations

QUESTION 6

Which type of PPE is used to protect the hands?

- a) Safety glasses
- b) Earplugs
- c) Gloves
- d) Steel-toed shoes

QUESTION 7

How should PPE be cleaned and maintained?

- a) With soap and water
- b) With chemicals
- c) With abrasive materials
- d) Not at all

QUESTION 8

What is the correct way to put on a respirator?

- a) By holding it in place with one hand
- b) By adjusting the straps to fit snugly
- c) By covering the mouth and nose with a cloth
- d) By not wearing it at all

QUESTION 9

Which type of PPE is used to protect the feet?

- a) Safety glasses
- b) Earplugs
- c) Gloves
- d) Steel-toed shoes

QUESTION 10

Why is it important to dispose of PPE properly?

- a) To avoid getting in trouble
- b) To protect the environment
- c) To comply with regulations
- d) To improve productivity

EMERGENCY PROCEDURES E.G., FIRE EVACUATION, FIRST AID

Fire Evacuation:

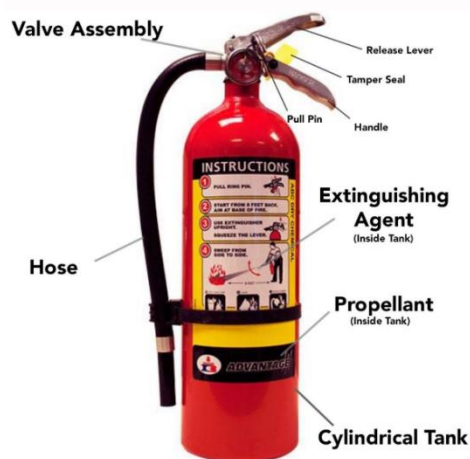
1. Know the location of fire alarms, extinguishers, and exits.
2. Understand the evacuation procedure:
 - Alert others and activate the alarm.
 - Evacuate the area via the nearest exit.
 - Assemble at the designated meeting point.
 - Do not use elevators.
3. Know how to use fire extinguishers:
 - Identify the type of extinguisher.

There are several types of fire extinguishers, each designed to combat specific classes of fires:

CLASSIFICATION	IMAGE	USE
Water Extinguishers (Class A)		➤ Effective against ordinary combustibles like paper, wood, and cloth. ➤ Not suitable for electrical or grease fires.
Carbon Dioxide (CO2) Extinguishers (Class B and C)		➤ Effective against flammable liquids and gases, as well as electrical fires. ➤ Not suitable for Class A fires.
Dry Chemical Extinguishers (Class B and C)		➤ Effective against flammable liquids, gases, and electrical fires. ➤ Not suitable for Class A fires.
Foam Extinguishers (Class B)		➤ Effective against flammable liquids and gases. ➤ Not suitable for electrical or Class A fires.
Wet Chemical Extinguishers (Class K)		➤ Effective against kitchen fires involving cooking oils and greases. ➤ Not suitable for other types of fires.

Halon Extinguishers (Class B and C)		➤ Effective against electrical and flammable liquid fires. ➤ Being phased out due to environmental concerns.
Clean Agent Extinguishers (Class B and C)		➤ Effective against electrical and flammable liquid fires. ➤ Environmentally friendly alternatives to Halon.
Cartridge-Operated Extinguishers (Class A, B, and C)		➤ Effective against various types of fires, depending on the cartridge. ➤ Often used in industrial settings.

Remember the PASS method to operate fire extinguishers:



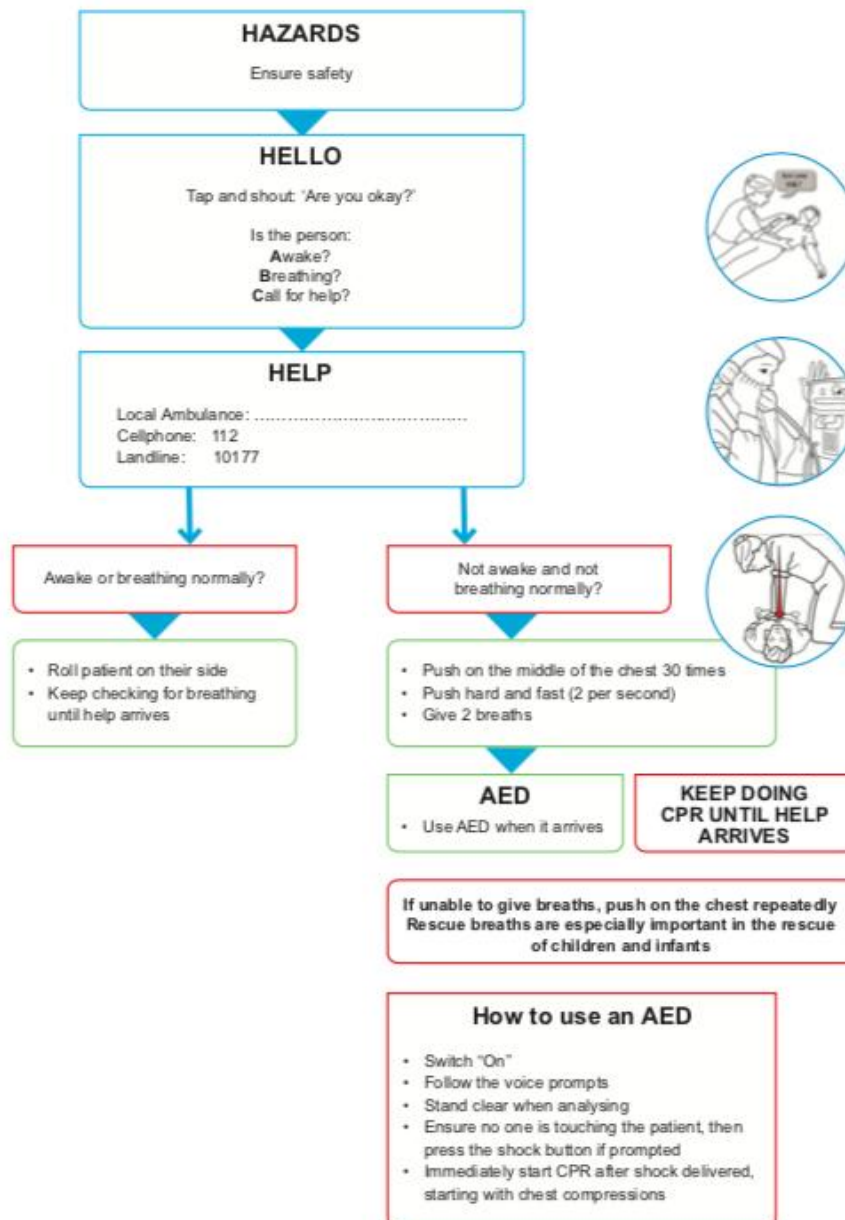
Follow the PASS method (Pull, Aim, Squeeze, Sweep).

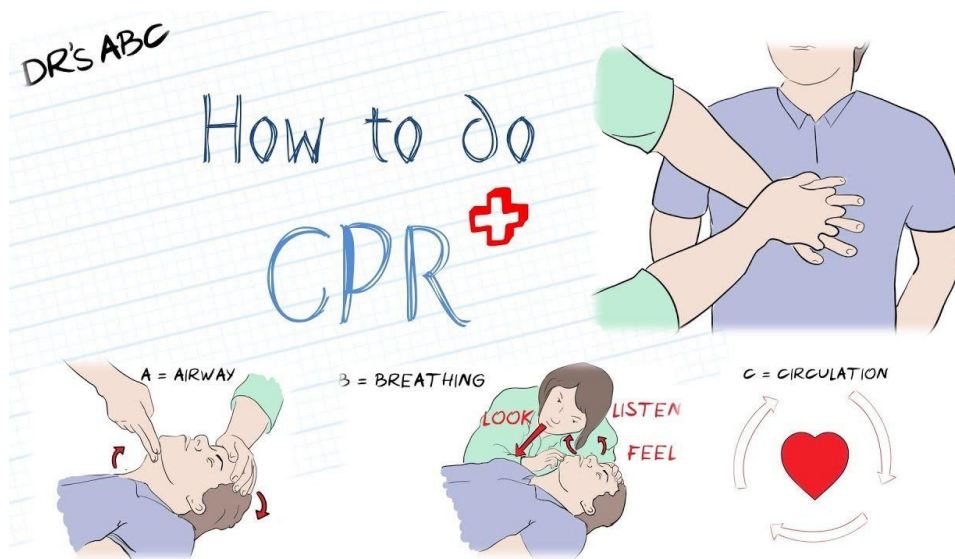
1. Pull the pin or ring.
2. Aim the nozzle or hose.
3. Squeeze the handle.
4. Sweep the extinguishing agent across the fire.

Always choose the correct extinguisher for the specific fire type and follow safety guidelines.

FIRST AID:

1. Know basic first aid techniques:
 - CPR (cardiopulmonary resuscitation).





WOUND CLEANING AND DRESSING

Wound cleaning and dressing are crucial steps in wound care to promote healing, prevent infection, and minimize scarring.



Preparation:

1. Wash your hands thoroughly with soap and water.
2. Put on gloves (latex) to prevent cross-contamination.
3. Prepare a clean workspace with necessary supplies.

Supplies needed:

- Saline solution (0.9% sodium chloride)
- Sterile gauze pads (2-3)
- Gauze roll or dressing pad
- Medical tape
- Antibiotic ointment (if prescribed)
- Gloves
- Scissors

Wound Cleaning:



1. Remove any debris or old dressings.
2. Rinse the wound with saline solution to loosen debris.
3. Use sterile gauze to gently clean the wound, working from the center outwards.
4. Avoid using harsh soap, hydrogen peroxide, or iodine, as they can damage tissue.
5. Repeat cleaning process until the wound is clear of debris.

Dressing Application:

1. Pat the wound dry with sterile gauze.
2. Apply antibiotic ointment (if prescribed) to the wound.
3. Cover the wound with a sterile gauze pad or dressing pad.
4. Secure the dressing with medical tape.
5. Use a gauze roll to reinforce the dressing, if necessary.

When to Seek Medical Attention:

1. Increased pain or discomfort
2. Excessive bleeding
3. Signs of infection (fever, redness, swelling, or discharge)
4. Delayed healing
5. Wound size increases

ACTIVITY 3

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Question 1

What is the primary purpose of wound dressing?

- A) To promote healing
- B) To prevent infection
- C) To protect the wound from further injury
- D) All of the above

Question 2

What type of wound dressing is suitable for a burn wound?

- A) Occlusive dressing
- B) Absorbent dressing
- C) Moisture-retentive dressing
- D) Hydrogel dressing

Question 3

What are the key principles of wound dressing selection?

- A) Cleanliness, sterility, and comfort
- B) Absorbency, breathability, and durability
- C) Adhesion, flexibility, and ease of removal
- D) All of the above

Question 4

What is the purpose of debridement in wound care?

- A) To promote healing
- B) To remove dead tissue
- C) To reduce bacterial load
- D) All of the above

Question 5

How often should wound dressings be changed?

- A) Daily
- B) Every 2-3 days
- C) Weekly
- D) As needed

Question 6

What is the first step in cleaning a wound?

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Question 7

What should be used to clean a wound?

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Question 8

What should be applied to a wound after cleaning?

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Question 9

Why is it important to clean a wound properly?

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Question 10

What happens if a wound is not cleaned properly?

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Question 11

What is the purpose of applying antibiotic ointment to a wound?

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Question 12

Demonstrate how to properly clean and dress a wound.

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Question 13

What should you do if you encounter a wound with heavy bleeding?

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Question 14

What are the signs of infection in a wound?

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Question 15

What should you do if you notice signs of infection in a wound?

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Question 16

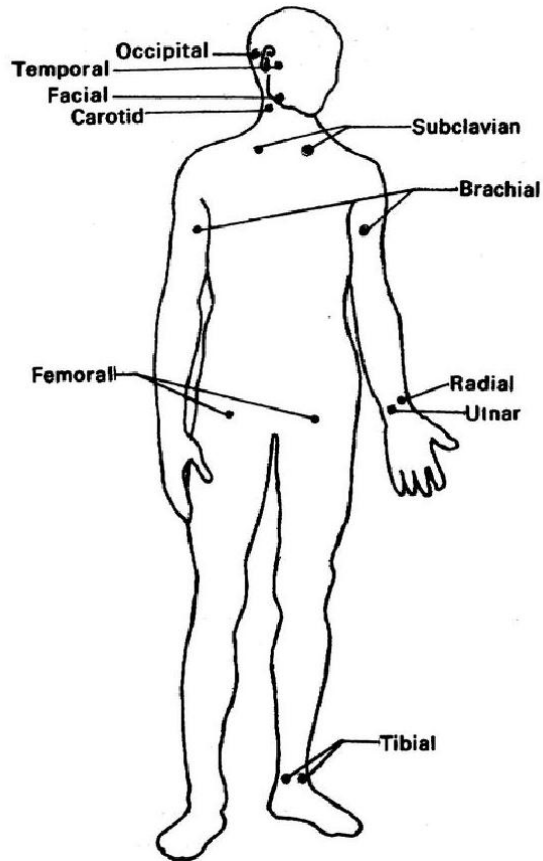
How can you determine if a wound is healing properly?

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BLEEDING CONTROL.
BLEEDING CONTROL TECHNIQUES

Control technique	method
Direct Pressure:- Apply direct pressure (5-10 minutes)	
Elevation	
Hemostatic Agents	
Cold applications	
Pressure bandage	
Pressure Points	



Common Bleeding Control Errors in South Africa

1. Inadequate direct pressure
2. Insufficient pressure duration
3. Failure to monitor vital signs

Burn care		
Classification of Burns	Affects	Treatment
First-degree (superficial):	Affects epidermis	Cool compresses - Topical creams - Pain management
Second-degree (partial-thickness):	Affects epidermis and dermis	Debridement - Antibiotic ointment - Dressing changes
Third-degree (full-thickness):	Affects all skin layers	Surgical excision - Skin grafting - Intravenous antibiotics
Fourth-degree:	Extends to muscle, bone, or organs	

ACTIVITY 4

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Fire Evacuation:

Question 1

What should you do in case of a fire?

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Question 2

Where should you assemble after evacuating a building during a fire?

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Question 3

What should you not use during a fire evacuation?

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Question 4

What is the first step in providing first aid for a cardiac emergency?

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Question 5

How should you clean a wound?

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Question 6

What should you do if someone is bleeding severely?

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Question 7

What should you do during an earthquake?

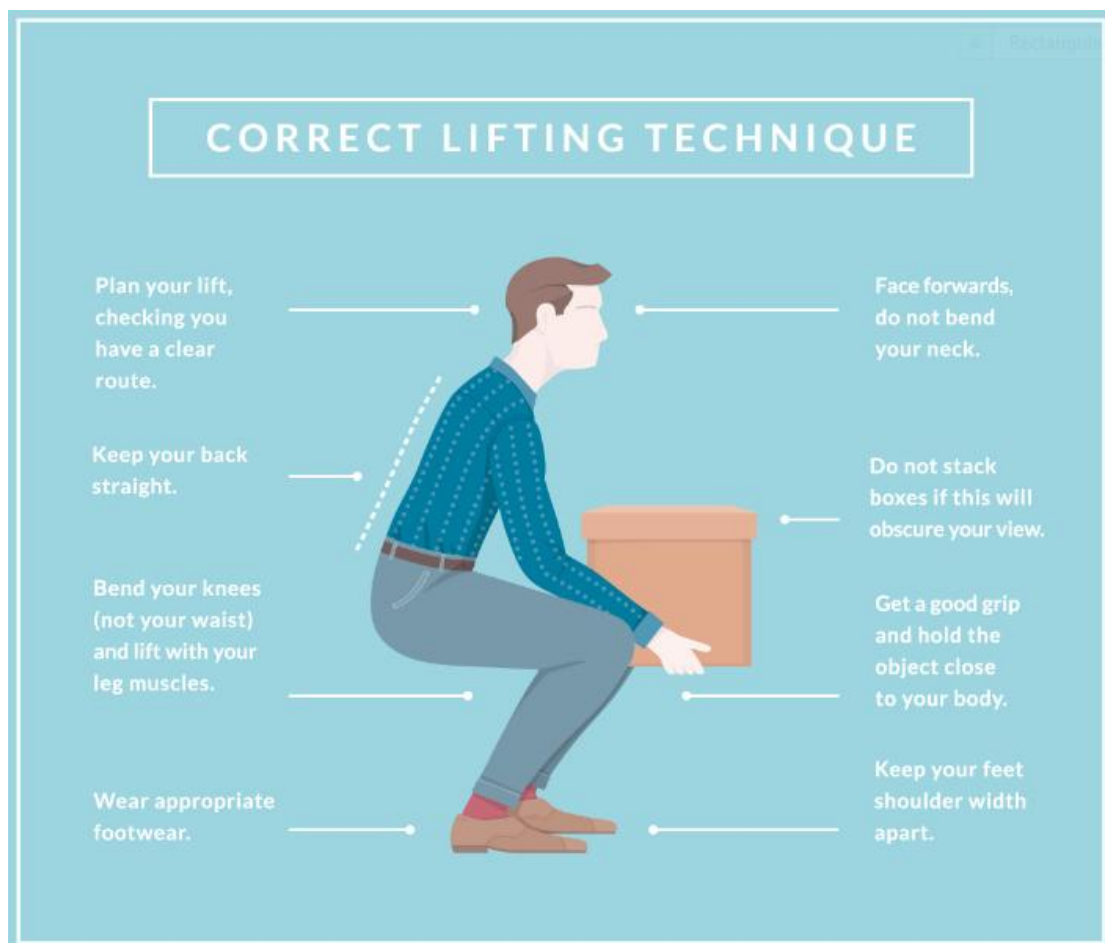
Question 8

What should you do in case of a chemical spill?

Question 9

What should you do during a power outage?

**EXPLAIN SAFE WORK PRACTICES
PROPER LIFTING TECHNIQUES**



1. Plan the lift: Consider the weight, size, and distance to be lifted.
2. Warm up: Stretch and flex muscles before lifting.
3. Position yourself: Stand close to the object, with feet shoulder-width apart.
4. Bend at the knees: Keep back straight, lift with legs, and avoid twisting.
5. Grip securely: Hold the object firmly, with hands at opposite sides.
6. Lift smoothly: Avoid jerky movements, lift steadily, and keep the object close to the body.
7. Set down carefully: Lower the object slowly and carefully, avoiding impact.

ACTIVITY 5

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Safe work practices (e.g., proper lifting techniques, tool use)

Question 1

What is the correct way to lift a heavy object?

- a) Bend at the waist and lift with your back
- b) Bend at the knees and lift with your legs
- c) Lift with your arms and twist your body
- d) Ask someone else to lift it for you

Question 2

Which of the following is a safe way to use a tool?

- a) Holding the tool with a loose grip
- b) Using the tool with a worn-out handle
- c) Keeping the tool at a comfortable distance
- d) Using the tool with a damaged blade

Question 3

What should you do before using a ladder?

- a) Inspect the ladder for damage
- b) Use the ladder without inspecting it
- c) Ask someone else to inspect the ladder
- d) Only use the ladder for short tasks

Question 4

What is the purpose of lockout / tagout procedures?

- a) To prevent equipment from being used
- b) To ensure equipment is used correctly
- c) To prevent unexpected startup of equipment
- d) To increase productivity

Question 5

What should you do if you notice a hazard in the work area?

- a) Ignore it and continue working
- b) Report it to your supervisor
- c) Try to fix it yourself
- d) Warn your coworkers

Question 6

What is the correct way to store tools?

- a) In a designated area, out of the way
- b) In a toolbox, without organization
- c) On the floor, in a pile
- d) In a locked cabinet, without access

Question 7

What should you wear when working with chemicals?

- a) Gloves and safety glasses
- b) A dust mask and earplugs
- c) A hard hat and steel-toed shoes
- d) A lab coat and closed-toe shoes

Question 8

What should you do if you are unsure about how to perform a task?

- a) Ask your supervisor for guidance
- b) Try to figure it out yourself
- c) Ask a coworker for help
- d) Skip the task and move on to the next one

DEMONSTRATE KNOWLEDGE OF SAFETY PROTOCOLS (E.G., LOCKOUT/TAGOUT, CONFINED SPACES).

Lockout/Tagout (LOTO)



1. Purpose: Prevent unexpected startup of equipment during maintenance or repair.



2. Steps:

- Notify affected employees and shut down equipment.
- Isolate energy sources (electrical, mechanical, hydraulic).
- Lock out energy sources with locks or tags.
- Verify zero energy state.

3. Importance: Prevents injuries and fatalities from unexpected startup.

Confined Spaces



1. Definition: Enclosed or partially enclosed spaces with limited entry/exit.

2. Hazards:

- Oxygen deficiency or enrichment.
- Toxic atmospheres.
- Explosive atmospheres.
- Physical hazards (entrapment, falling).

3. Safety protocols:

- Permit-required confined spaces: Obtain permit, follow entry procedures.
- Non-permit confined spaces: Follow entry procedures, monitor atmosphere.
- Entry procedures: Test atmosphere, ventilate, use PPE, maintain communication.

ACTIVITY 6

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

SAFETY PROTOCOLS (E.G., LOCKOUT/TAGOUT, CONFINED SPACES)

Question 1

An employee needs to perform maintenance on a machine. What should they do first?

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Question 2

A worker enters a confined space without a permit. What is the risk?

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Question 3

A hot work permit is required for a welding task. What should the worker do?

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Question 4

What is the purpose of lockout/tagout procedures?

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Question 5

What are the steps for lockout/tagout?

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Question 6

Why is lockout/tagout important?

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Question 7

What is a confined space?

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Question 8

What are the hazards associated with confined spaces?

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Question 9

What are the safety protocols for confined spaces?

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.....

HIV and AIDS Procedures:

Demonstrate understanding of HIV and AIDS transmission and prevention.

HIV and AIDS transmission and prevention.

HIV Transmission:

1. Sexual transmission: unprotected sex (vaginal, anal, oral)
2. Blood transmission: blood transfusions, sharing needles, mother-to-child
3. Bodily fluids: breast milk, semen, vaginal fluids, blood

HIV Prevention:

1. Safe sex practices: condoms, monogamy
2. Needle exchange programs
3. Blood screening: donor screening, safe blood transfusions
4. Antiretroviral therapy (ART): pre-exposure prophylaxis (PrEP), post-exposure prophylaxis (PEP)
5. Mother-to-child prevention: ART during pregnancy, cesarean delivery, formula feeding

AIDS Prevention:

1. Early HIV diagnosis and treatment
2. ART adherence
3. Healthy lifestyle: nutrition, exercise, stress management
4. Opportunistic infection prevention: PCP, TB, pneumonia vaccines

Key Terms:

1. HIV: Human Immunodeficiency Virus
2. AIDS: Acquired Immune Deficiency Syndrome
3. ART: Antiretroviral Therapy
4. PrEP: Pre-Exposure Prophylaxis
5. PEP: Post-Exposure Prophylaxis

AIDS is the most advanced stage of HIV infection

ACTIVITY 7

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

HIV and AIDS transmission and prevention:

Question 1

What is the primary mode of HIV transmission?

- A) Blood transfusions
- B) Unprotected sex
- C) Mother-to-child transmission
- D) Casual contact

Question 2

Which of the following is an effective method for preventing HIV transmission?

- A) Using condoms
- B) Sharing needles
- C) Blood transfusions without screening
- D) Not using condoms

Question 3

What is the purpose of antiretroviral therapy (ART) in HIV prevention?

- A) Cure HIV infection
- B) Prevent HIV transmission
- C) Treat opportunistic infections
- D) Reduce viral load

Question 4

What is the benefit of pre-exposure prophylaxis (PrEP) in HIV prevention?

- A) Treats HIV infection
- B) Prevents HIV transmission
- C) Cures AIDS
- D) Reduces ART effectiveness

Question 5

According to the World Health Organization (WHO), what is the estimated number of people living with HIV globally?

- A) 10 million
- B) 20 million
- C) 38 million
- D) 50 million

Question 6

What is the window period for HIV testing?

- A) 1-3 months
- B) 3-6 months
- C) 6-12 months
- D) 1-2 years

Question 7

Which of the following groups is most vulnerable to HIV infection?

- A) Pregnant women
- B) Sex workers
- C) Men who have sex with men (MSM)
- D) All of the above

Question 8

What is the purpose of post-exposure prophylaxis (PEP)?

- A) Treat HIV infection
- B) Prevent HIV transmission after exposure
- C) Cure AIDS
- D) Reduce ART effectiveness

Question 9

How can mother-to-child transmission of HIV be prevented?

- A) ART during pregnancy
- B) Cesarean delivery
- C) Formula feeding
- D) All of the above

Question 10

What is the benefit of voluntary counseling and testing (VCT) for HIV?

- A) Increases HIV transmission
- B) Decreases HIV testing
- C) Improves treatment adherence
- D) Reduces stigma

1. SHOW KNOWLEDGE OF UNIVERSAL PRECAUTIONS (E.G., GLOVE USE, HAND HYGIENE).

Definition:

Universal precautions are infection control measures used to prevent transmission of pathogens in healthcare settings.

Key Principles:

1. Treat all patients as potentially infectious.
2. Use personal protective equipment (PPE) consistently.
3. Follow proper hand hygiene practices.

Universal Precautions:-

1. Glove Use:

- Wear gloves for all patient contacts.
- Change gloves between patients.
- Use sterile gloves for invasive procedures.

2. Hand Hygiene:

- Wash hands with soap and water.
- Use alcohol-based hand rubs.
- Perform hand hygiene before/after patient contact.

3. Personal Protective Equipment (PPE):

- Wear masks, eye protection, and gowns as needed.
- Use respirators for airborne pathogens.

4. Needle Safety:

- Use needle-free devices when possible.
- Dispose of needles safely.

5. Surface Cleaning:

- Clean and disinfect surfaces regularly.

Key Terms:

1. PPE (Personal Protective Equipment)
2. Hand hygiene

- 3. Sterilization
- 4. Disinfection
- 5. Infection control

ACTIVITY 8

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Universal Precautions:

Question 1

What is the primary purpose of universal precautions?

- A) Prevent patient-to-patient transmission
- B) Prevent healthcare worker-to-patient transmission
- C) Prevent transmission of all pathogens
- D) Reduce hospital-acquired infections

Question 2

Which of the following is an essential component of universal precautions?

- A) Glove use
- B) Hand hygiene
- C) Mask use
- D) All of the above

Question 3

When should gloves be worn?

- A) Only during invasive procedures
- B) During all patient contacts
- C) When handling bodily fluids
- D) When handling medications

Question 4

What is the recommended hand hygiene practice?

- A) Wash hands with soap and water
- B) Use alcohol-based hand rubs
- C) Both A and B
- D) Neither A nor B

Question 5

What type of mask should be worn for airborne pathogens?

- A) Surgical mask
- B) Respirator mask

- C) Dust mask
- D) Face shield

Question 6

How often should gloves be changed?

- A) Between patients
- B) After each procedure
- C) Every hour
- D) At the end of the shift

Question 7

What is the purpose of surface cleaning?

- A) Reduce hospital-acquired infections
- B) Prevent transmission of pathogens
- C) Maintain hospital aesthetics
- D) Both A and B

Question 8

Which of the following is an example of a sharps injury?

- A) Needle stick
- B) Scalpel cut
- C) Glass shard injury
- D) All of the above

Question 9

What should be done with used needles?

- A) Recap them
- B) Bend them
- C) Dispose of them safely
- D) Reuse them

Question 10

Who is responsible for implementing universal precautions?

- A) Healthcare workers
- B) Patients
- C) Family members
- D) All of the above

EXPLAIN POST-EXPOSURE PROPHYLAXIS (PEP) PROCEDURES.

POST-EXPOSURE PROPHYLAXIS (PEP) PROCEDURES:

Definition: PEP is an emergency medication taken after potential exposure to HIV, hepatitis, or other bloodborne pathogens to prevent infection.

PEP Procedures:

(a) Assessment

1. Evaluate exposure risk (e.g., needle stick, sexual assault)
2. Determine exposure source (e.g., HIV-positive individual)
3. Assess exposure severity (e.g., deep wound, high-risk fluid)

(b) Initiation

1. Start PEP within 72 hours (ideally within 2 hours) of exposure
2. Consult occupational health or HIV specialist
3. Obtain informed consent

(c) Medication Regimen

1. HIV PEP: Antiretroviral therapy (ART) combination (e.g., tenofovir/emtricitabine)
2. Hepatitis PEP: Hepatitis B immune globulin (HBIG) and/or hepatitis B vaccine
3. Duration: Typically 28 days

(d) Follow-up

1. Monitor for side effects and adherence
2. Conduct HIV testing at baseline, 6 weeks, 12 weeks, and 6 months
3. Provide counseling and support

(e) Documentation

1. Record exposure details and PEP initiation
2. Monitor medication adherence and side effects
3. Document follow-up appointments and test results

Key Terms:

1. PEP (Post-Exposure Prophylaxis)
2. ART (Antiretroviral Therapy)
3. HBIG (Hepatitis B Immune Globulin)
4. Baseline testing

ACTIVITY 9

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Post-Exposure Prophylaxis (PEP):

Question 1

What is the primary purpose of PEP?

- A) Treat HIV infection
- B) Prevent HIV transmission
- C) Cure AIDS
- D) Reduce ART effectiveness

Question 2

Within what time frame should PEP be initiated?

- A) 24 hours
- B) 48 hours
- C) 72 hours
- D) 1 week

Question 3

What is the typical duration of PEP medication?

- A) 14 days
- B) 28 days
- C) 30 days
- D) 60 days

Question 4

Which of the following is a common PEP medication regimen?

- A) Tenofovir/emtricitabine
- B) Zidovudine/lamivudine
- C) Efavirenz/tenofovir
- D) All of the above

Question 5

What is the purpose of baseline HIV testing in PEP?

- A) Monitor treatment efficacy
- B) Determine HIV status
- C) Assess exposure risk
- D) Evaluate medication adherence

Question 6

How often should HIV testing be conducted during PEP?

- A) Weekly
- B) Bi-weekly
- C) Monthly
- D) At baseline, 6 weeks, 12 weeks, and 6 months

Question 7

What is the role of HBIG in PEP?

- A) Prevents HIV transmission
- B) Treats hepatitis B infection
- C) Prevents hepatitis B transmission
- D) Boosts immune system

Question 8

Who should receive PEP after a needle stick injury?

- A) Only healthcare workers
- B) All individuals exposed
- C) Only those with high-risk exposure
- D) None of the above

Question 9

What is the importance of adherence to PEP medication?

- A) Ensures treatment efficacy
- B) Reduces side effects
- C) Prevents resistance
- D) All of the above

Question 10

Which organization provides guidelines for PEP?

- A) WHO
- B) CDC
- C) South African National Department of Health
- D) All of the above

2. DEMONSTRATE UNDERSTANDING OF HIV AND AIDS CONFIDENTIALITY AND DISCLOSURE PROTOCOLS.

HIV and AIDS Confidentiality and Disclosure Protocols: Confidentiality Protocols:

1. Maintain patient confidentiality.
2. Protect HIV-related information.
3. Limit access to authorized personnel.
4. Use secure records and storage.
5. Ensure informed consent for disclosure.

Disclosure Protocols:

1. Obtain informed consent.
2. Disclose to authorized personnel (e.g., healthcare providers).
3. Disclose to partners/spouses (contact tracing).
4. Disclose in court/legal proceedings (subpoena).
5. Disclose in cases of child/elder abuse.

Key Terms:

1. Confidentiality
2. Disclosure
3. Informed consent
4. Authorized personnel
5. Contact tracing

Confidentiality Breach Consequences:

1. Legal action
2. Professional disciplinary action
3. Patient mistrust
4. Social stigma

Disclosure Benefits:

1. Prevents transmission
2. Promotes treatment adherence
3. Enhances partner testing
4. Supports public health interventions

ACTIVITY 10

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

HIV and AIDS Confidentiality and Disclosure Protocols:

Question 1

What is the primary purpose of HIV confidentiality protocols?

- A) Protect patient identity
- B) Prevent transmission
- C) Promote treatment adherence
- D) Ensure public health safety

Question 2

Which of the following is authorized to access HIV-related information?

- A) Healthcare providers
- B) Family members
- C) Employers
- D) Law enforcement

Question 3

What is required for disclosure of HIV status?

- A) Informed consent
- B) Verbal consent
- C) Written consent
- D) Implied consent

Question 4

Who can disclose HIV information in cases of child/elder abuse?

- A) Healthcare providers
- B) Social workers
- C) Law enforcement
- D) All of the above

Question 5

What is contact tracing in HIV disclosure?

- A) Informing partners/spouses of HIV status
- B) Notifying public health authorities
- C) Conducting HIV testing
- D) Providing treatment

Question 6

What is the consequence of breaching HIV confidentiality?

- A) Legal action
- B) Professional disciplinary action
- C) Patient mistrust
- D) All of the above

Question 7

Which international organization provides guidelines on HIV confidentiality?

- A) WHO
- B) UNAIDS
- C) CDC
- D) All of the above

Question 8

What is the purpose of the National Health Act (2003) in South Africa?

- A) Regulate healthcare services
- B) Protect patient confidentiality
- C) Promote public health

D) All of the above

Question 9

When can HIV information be disclosed in court/legal proceedings?

- A) With informed consent
- B) With subpoena
- C) With court order
- D) All of the above

Question 10

What is the benefit of maintaining HIV confidentiality?

- A) Prevents transmission
- B) Promotes treatment adherence
- C) Reduces stigma
- D) All of the above

SHOW KNOWLEDGE OF HIV AND AIDS TESTING AND COUNSELING PROCEDURES.

HIV AND AIDS TESTING AND COUNSELING PROCEDURES

Testing Procedures:

1. Informed Consent: Obtain written consent before testing.
2. Pre-Test Counseling: Discuss risks, benefits, and implications.
3. Blood Sample Collection: Venous or finger-prick sample.
4. Rapid Testing: Results within 15-30 minutes.
5. Laboratory Testing: ELISA, Western Blot, or PCR.
6. Post-Test Counseling: Interpret results, discuss implications.

Counseling Procedures:

1. Pre-Test Counseling:
 - Assess risk factors
 - Discuss testing benefits
 - Address concerns
2. Post-Test Counseling (Positive Result):
 - Emotional support
 - Treatment options
 - Disclosure guidance
 - Partner notification
3. Post-Test Counseling (Negative Result):
 - Risk reduction strategies
 - Prevention methods
 - Repeat testing recommendations

Types of HIV Tests:

1. Rapid Tests (e.g., OraQuick)
2. ELISA (Enzyme-Linked Immunosorbent Assay)
3. Western Blot
4. PCR (Polymerase Chain Reaction)
5. Viral Load Testing

Key Terms:

1. Informed consent
2. Pre-test counseling
3. Post-test counseling
4. Rapid testing
5. Viral load testing

Benefits of HIV Testing:

1. Early diagnosis
2. Timely treatment
3. Reduced transmission
4. Improved health outcomes

Challenges:

1. Stigma and discrimination
2. Limited access to testing
3. Delayed diagnosis
4. Linkage to care

ACTIVITY 11**KWA-THEMA SKILLS SCHOOL**

Learner Name:..... Surname:

Term : 4 Date :

HIV and AIDS Testing and Counseling Procedures

Question 1

What is the primary purpose of pre-test counseling?

- A) To obtain informed consent
- B) To discuss testing benefits
- C) To assess risk factors
- D) All of the above

Question 2

Which type of HIV test provides results within 15-30 minutes?

- A) ELISA
- B) Western Blot
- C) Rapid test
- D) PCR

Question 3

What is the benefit of provider-initiated testing?

- A) Increased patient autonomy
- B) Reduced stigma
- C) Early diagnosis and treatment
- D) Decreased testing costs

Question 4

What is the purpose of post-test counseling for a positive result?

- A) To discuss treatment options
- B) To provide emotional support
- C) To guide disclosure
- D) All of the above

Question 5

Which of the following is a type of HIV test?

- A) OraQuick
- B) ELISA
- C) Western Blot
- D) All of the above

Question 6

What is the recommended testing frequency for high-risk individuals?

- A) Every 3 months
- B) Every 6 months
- C) Every 12 months
- D) As needed

Question 7

What is the purpose of informed consent in HIV testing?

- A) To ensure patient understanding
- B) To protect healthcare provider liability
- C) To maintain confidentiality

D) All of the above

Question 8

Which organization provides guidelines for HIV testing?

- A) WHO
- B) CDC
- C) South African National Department of Health
- D) All of the above

Question 9

What is the benefit of couple testing?

- A) Increased testing accuracy
- B) Improved relationship dynamics
- C) Enhanced prevention strategies
- D) Reduced stigma

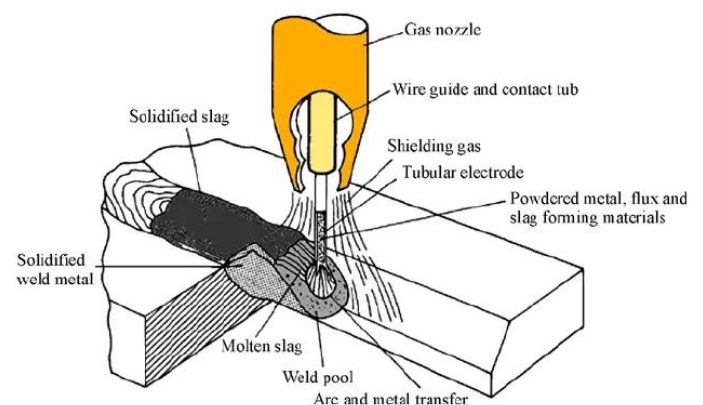
Question 10

What is the purpose of viral load testing?

- A) Diagnose HIV infection
- B) Monitor treatment effectiveness
- C) Detect resistance
- D) All of the above

CHAPTER 2 & 3

CORED- WIRE WELDING



Cored-wire welding process.

Select, set up and conduct pre-operational checks of cored-wire welding equipment.

Cored-wire welding, also known as flux-cored arc welding (FCAW), is a semi-automatic welding process that uses a continuous wire electrode with a flux core.

The process combines the benefits of shielded metal arc welding (SMAW) and gas metal arc welding (GMAW).

Equipment and Materials

1. Welding machines (DC or AC)
2. Cored-wire (various diameters and types)
3. Shielding gases (e.g., CO₂, Ar+CO₂)
4. Welding helmets and protective gear

Equipment Setup:

1. Welding power source
2. Wire feed unit
3. Gun/cable assembly
4. Shielding gas supply
5. Welding helmet

What are pre-operational checks before welding?

This is carried out in accordance with manufacturer's specifications.

Prepare work pieces prior to welding.

- 1) Work pieces prepared prior to welding as specified on drawing and worksite procedures.
- 2) Dimensions and alignment checked as specified on drawing.
- 3) Workpiece tack welded in position as per drawing specifications.

Equipment Inspection:

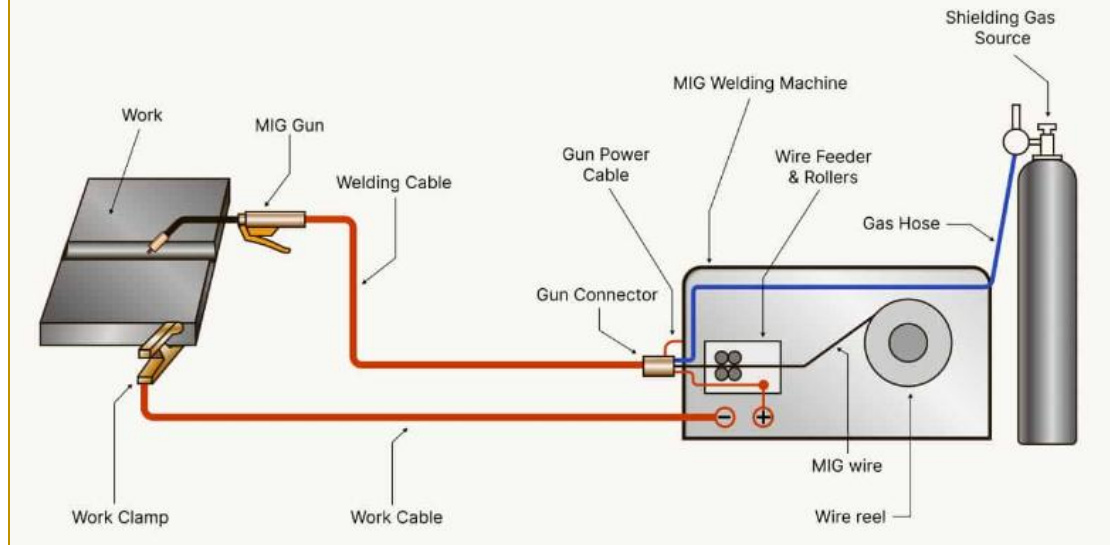
1. Power source: Ensure proper voltage, amperage, and polarity.
2. Wire feed unit: Check wire feed speed, tension, and alignment.
3. Gun/cable assembly: Inspect for damage, wear, or corrosion.
4. Shielding gas supply: Verify gas flow rate and cylinder pressure.

Safety Checks:

1. Personal protective equipment (PPE)
2. Ventilation and fire safety
3. Electrical safety

Setup Procedures:

MIG Welding Setup



MIG Welding Operating Parameters: Understanding Welding Gases

Material Thickness (mm)	Wire Diameter (mm)	Voltage (V)	Current (A)
0.5	0.6	15-17	20-30
1.0	0.6-0.8	16-18	30-40
1.5	0.6-0.8	17-19	35-50
2.0	0.8-1.0	18-20	40-90
2.5	0.8-1.0	19-21	50-100
3.0	0.8-1.2	20-22	90-120
3.5	0.8-1.2	21-23	100-130
4.0	0.8-1.2	22-24	120-150
4.5	0.8-1.2	23-25	140-170
5.0	0.8-1.2	24-26	150-180
5.5	0.8-1.2	25-27	160-190
6.0	0.8-1.2	26-28	180-210

Welding Techniques:

The typical wire feed speed range for Flux Cored Arc Welding (FCAW) can vary depending on several factors, including:

1. Wire diameter

2. Metal thickness
3. Welding position
4. Shielding gas
5. Equipment specifications

However, here are some general guidelines for wire feed speed ranges for FCAW in South Africa:

FCAW Wire Feed Speed Ranges (mm/s):

- 0.8-1.2 mm wire diameter: 4-8 mm/s (e.g., for thin materials)
- 1.0-1.4 mm wire diameter: 5-10 mm/s (e.g., for medium-thickness materials)
- 1.2-1.6 mm wire diameter: 6-12 mm/s (e.g., for thick materials)

Specific Wire Feed Speed Ranges for Common FCAW Applications:

- Structural steel fabrication: 5-9 mm/s
- Shipbuilding: 6-12 mm/s
- Pressure vessel fabrication: 4-8 mm/s
- Pipe welding: 5-10 mm/s

South African Standards and Guidelines:

- South African National Standards (SANS) 1597:2013 - Welding and allied processes - Flux cored arc welding
- South African Institute of Welding (SAIW) - FCAW Guidelines

International Standards and Guidelines:

- American Welding Society (AWS) A5.29/A5.29M - Specification for Flux Cored Arc Welding
- International Organization for Standardization (ISO) 17632 - Welding - Flux cored arc welding

NB:-Please note that these are general guidelines, and optimal wire feed speeds may vary depending on specific welding conditions and equipment.

ACTIVITY 12

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Flux Cored Arc Welding (FCAW) Wire Feed Speed:

Question 1

What is the primary factor affecting FCAW wire feed speed?

- A) Wire diameter
- B) Metal thickness
- C) Welding position
- D) Shielding gas

Question 2

What is the typical wire feed speed range for 1.0-1.4 mm wire diameter?

- A) 4-8 mm/s
- B) 5-10 mm/s
- C) 6-12 mm/s
- D) 8-15 mm/s

Question 3

What is the recommended wire feed speed for structural steel fabrication?

- A) 4-8 mm/s
- B) 5-9 mm/s
- C) 6-12 mm/s
- D) 8-15 mm/s

Question 4

Which standard provides guidelines for FCAW wire feed speeds in South Africa?

- A) SANS 1597:2013
- B) AWS A5.29/A5.29M
- C) ISO 17632
- D) All of the above

Question 5

What is the effect of increasing wire feed speed on weld penetration?

- A) Decreases penetration
- B) Increases penetration
- C) No effect on penetration
- D) Depends on wire diameter

Question 6

What is the optimal wire feed speed for pipe welding?

- A) 4-8 mm/s
- B) 5-10 mm/s
- C) 6-12 mm/s
- D) 8-15 mm/s

Question 7

How does shielding gas affect FCAW wire feed speed?

- A) Increases speed
- B) Decreases speed
- C) No effect
- D) Depends on gas composition

Question 8

What is the consequence of incorrect wire feed speed?

- A) Porosity
- B) Lack of fusion
- C) Overheating
- D) All of the above

Question 9

Which welding position requires slower wire feed speeds?

- A) Flat
- B) Horizontal
- C) Vertical
- D) Overhead

Question 10

What is the benefit of optimizing wire feed speed in FCAW?

- A) Increased productivity
- B) Improved weld quality
- C) Reduced costs
- D) All of the above

STEP-BY-STEP GUIDE WITH PICTURES TO EXPLAIN THE IMPORTANCE OF CORRECT ASSEMBLY OF CORED-WIRE WELDING EQUIPMENT:

Correct Assembly:

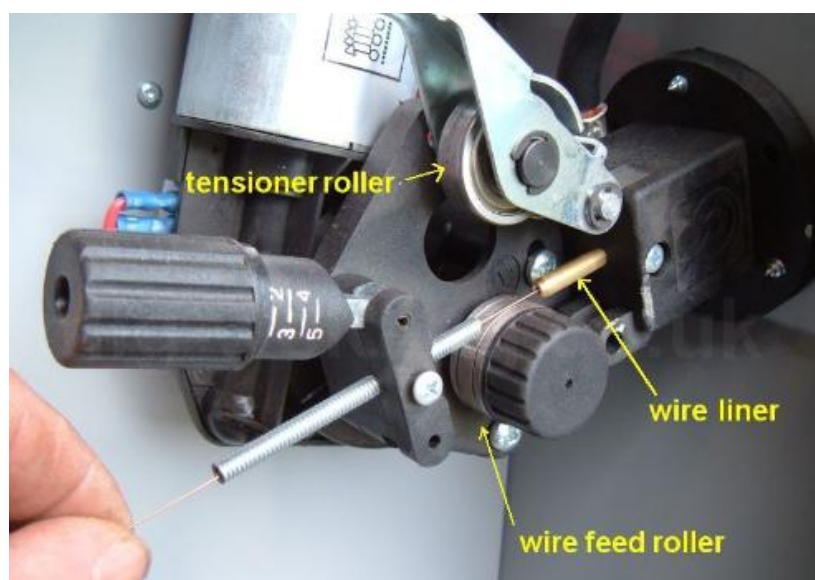


Step 1: Power Source Connection

- Connect power source to wire feed unit, ensuring secure and clean connections.
- Refer to vendor manual for specific connection requirements.

Step 2: Wire Feed Unit Setup

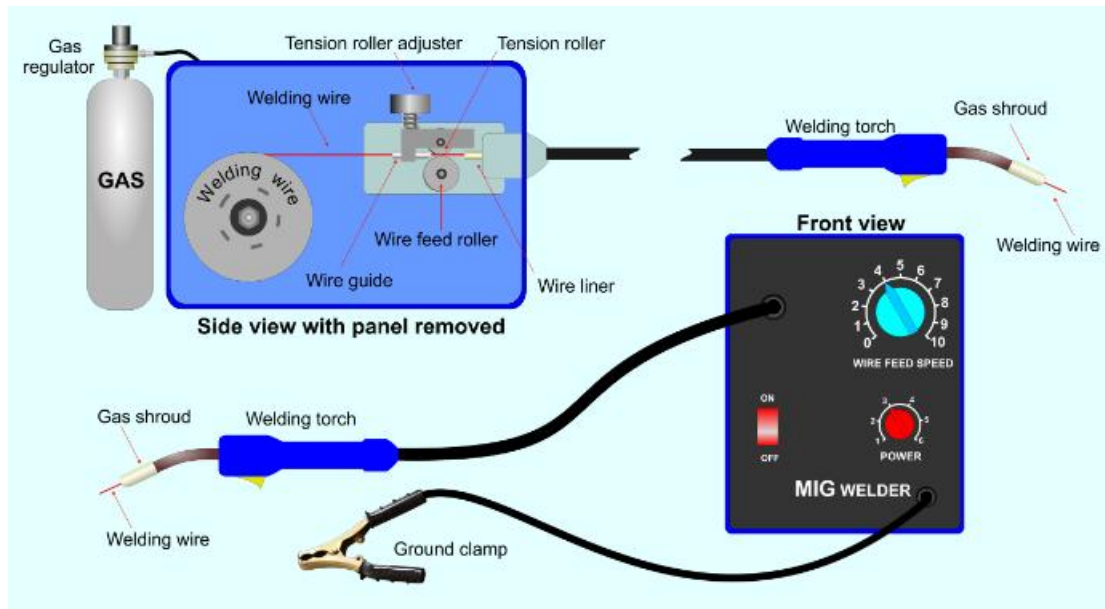
Setting up wire feed unit



- Set wire feed speed and tension according to vendor recommendations.
- Ensure wire feed unit is properly aligned with gun.

Step 3: Gun/Cable Assembly

Connecting gun to wire feed unit



- Connect gun to wire feed unit, ensuring secure connections.
- Check gun cable for damage or wear.

Step 4: Shielding Gas Connection

Connecting shielding gas cylinder



- Connect shielding gas cylinder to regulator.
- Set gas flow rate according to vendor recommendations.

Step 5: Wire Electrode Installation

- Install wire electrode according to vendor instructions.

- Ensure wire electrode is properly seated.

CONSEQUENCES OF INCORRECT ASSEMBLY

Incorrectly assembled cored-wire welding equipment

- Poor wire feedability
- Insufficient shielding gas flow
- Electrical shock or fire hazards
- Reduced weld quality
- Equipment damage

ACTIVITY 13

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Correct Assembly of Cored-Wire Welding Equipment

1. What is the primary purpose of following vendor requirements for cored-wire welding equipment assembly?

- A) To ensure safety
- B) To improve weld quality
- C) To increase productivity
- D) To reduce costs

2. What is the consequence of incorrect wire feed unit setup?

- A) Improved weld penetration
- B) Reduced wire feedability
- C) Increased shielding gas flow
- D) Enhanced safety

3. What should be checked before connecting the power source to the wire feed unit?

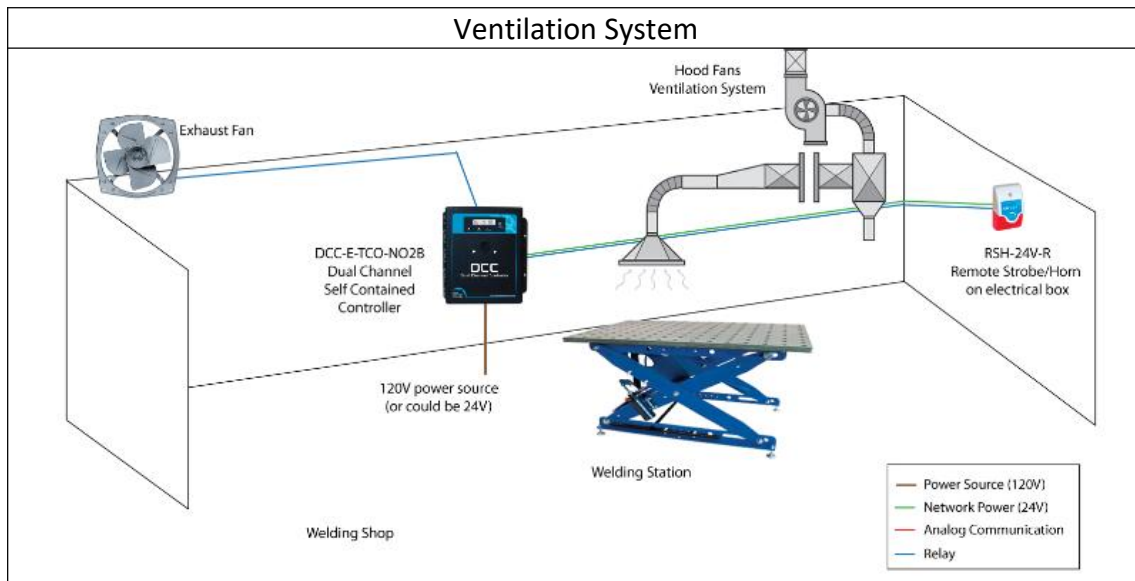
- A) Wire electrode installation
- B) Shielding gas flow rate
- C) Clean connections
- D) Gun cable damage

4. What is the result of insufficient shielding gas flow?
- A) Improved weld quality
 - B) Reduced porosity
 - C) Increased weld penetration
 - D) Porosity and lack of fusion
5. What is the purpose of setting the correct wire feed speed?
- A) To improve weld penetration
 - B) To reduce wire feedability
 - C) To ensure consistent weld quality
 - D) To increase productivity
6. What is the consequence of incorrect gun/cable assembly?
- A) Improved weld quality
 - B) Reduced electrical shock risk
 - C) Increased wire feedability
 - D) Electrical shock or fire hazards
7. What should be consulted for specific assembly requirements?
- A) Vendor manual
 - B) Safety data sheets (SDS)
 - C) Welding equipment manufacturer's instructions
 - D) All of the above
8. What is the result of incorrect wire electrode installation?
- A) Improved weld quality
 - B) Reduced wire feedability
 - C) Increased weld penetration
 - D) Wire feed problems and equipment damage
9. Why is proper grounding important in cored-wire welding equipment assembly?
- A) To improve weld quality
 - B) To reduce electrical shock risk
 - C) To increase productivity
 - D) To reduce costs
10. What should be done after assembling cored-wire welding equipment?
- A) Begin welding immediately
 - B) Conduct safety checks
 - C) Consult vendor manual

D) Perform equipment maintenance

SHIELDING GAS SUPPLY COMPONENTS:		
1.	Gas Cylinder	
2.	Regulator	
3.	Gas Hose	
4.	Gas Flow Meter	
Wire Electrode Components:		
5.	Core wire	

6.	Flux Core	
7.	Wire Sheath	
8.	Welding Wire	
Other Components:		
9.	Welding Helmet	
10.	Welding Gloves	
11.	Welding Jacket	
12.	Safety Glasses	



ACTIVITY 14

KWA-THEMA SKILLS SCHOOL

Learner Name: Surname:

Term : 4 Date :

Cored-Wire Welding Components:

1. What is the primary function of the power source in cored-wire welding?

- A) To generate shielding gas
- B) To feed wire electrode
- C) To provide electrical energy
- D) To regulate wire feed speed

2. What component feeds the wire electrode in cored-wire welding?

- A) Gun/Cable Assembly
- B) Wire Feed Unit
- C) Power Source
- D) Shielding Gas Supply

Answer: B) Wire Feed Unit

3. What is the purpose of the shielding gas diffuser in cored-wire welding?

- A) To protect the arc

- B) To regulate wire feed speed
- C) To generate heat
- D) To cool the wire electrode

4. What type of wire electrode is used in cored-wire welding?

- A) Solid wire
- B) Flux-cored wire
- C) Metal-cored wire
- D) All of the above

5. What component controls the wire feed speed in cored-wire welding?

- A) Wire Feed Motor
- B) Wire Feed Rolls
- C) Wire Tension Adjuster
- D) Wire Feed Speed Control

6. What is the purpose of the contact tip in cored-wire welding?

- A) To generate shielding gas
- B) To feed wire electrode
- C) To maintain electrical contact
- D) To regulate wire feed speed

7. What component protects the welder from electrical shock?

- A) Welding Helmet
- B) Welding Gloves
- C) Grounding Cable
- D) Safety Glasses

8. What is the purpose of the gas regulator in cored-wire welding?

- A) To generate shielding gas
- B) To regulate gas flow rate
- C) To maintain gas pressure
- D) To filter gas

9. What component holds the wire electrode in place?

- A) Wire Electrode Holder
- B) Gun/Cable Assembly
- C) Wire Feed Unit
- D) Power Source

10. What is the purpose of the ventilation system in cored-wire welding?

- A) To generate shielding gas
- B) To regulate wire feed speed
- C) To remove fumes and particles

D) To maintain gas pressure

WELDING TERMINOLOGY AS RECORDED IN INTERNATIONAL WELDING STANDARDS.

INTERNATIONAL WELDING STANDARDS:-

1. AWS (American Welding Society)
2. ISO (International Organization for Standardization)
3. ASME (American Society of Mechanical Engineers)

Welding Terminology:

1. Weld: A joint made by fusing two or more workpieces.
2. Welding: A fabrication process joining materials by heat, pressure, or both.
3. Fusion Welding: Welding process melting base metal.
4. Solid-State Welding: Welding process without melting base metal.

Cored-Wire Welding Terms:

1. Cored-Wire: Tubular wire electrode with flux core.
2. Flux-Cored Arc Welding (FCAW): Welding process using cored-wire.
3. Shielding Gas: Gas protecting arc and weld area.
4. Wire Feed Speed: Speed wire electrode fed through gun.

Welding Processes:

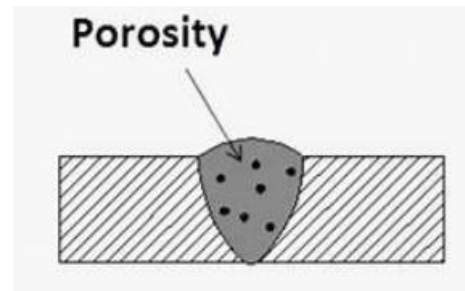
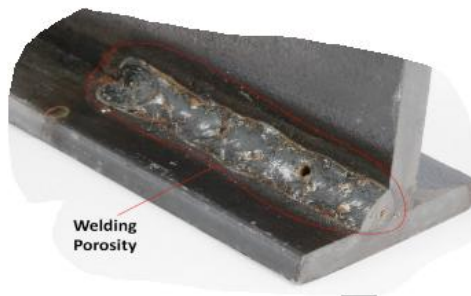
1. Gas Metal Arc Welding (GMAW)
2. Gas Tungsten Arc Welding (GTAW)
3. Shielded Metal Arc Welding (SMAW)
4. Flux-Cored Arc Welding (FCAW)

Welding Defects:

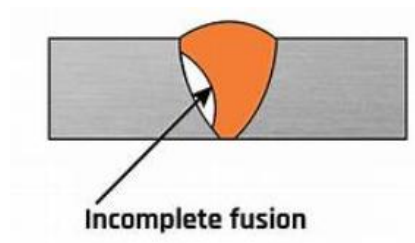
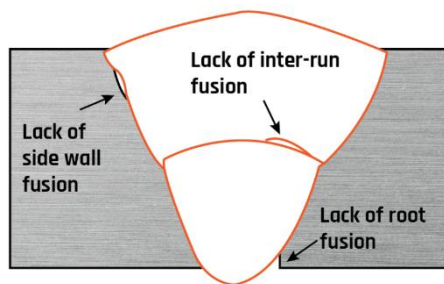
1. Porosity: Cavities or pores in weld.

Identification

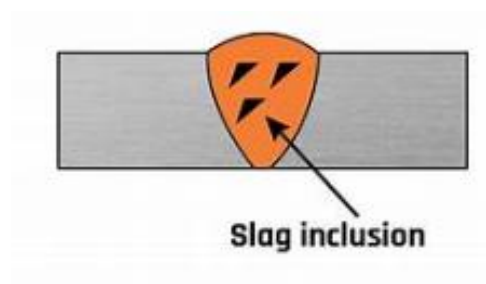
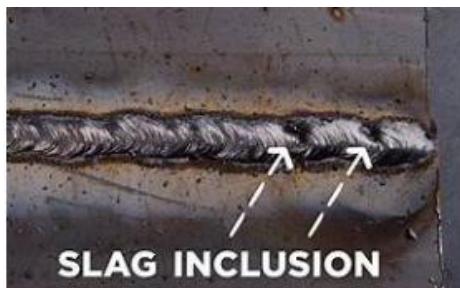
- Porosity is the presence of cavities in the weld metal caused by the freezing in of gas released from the weld pool as it solidifies.
- The porosity can take several forms:
 - distributed
 - surface breaking pores
 - wormhole
 - crater pipes



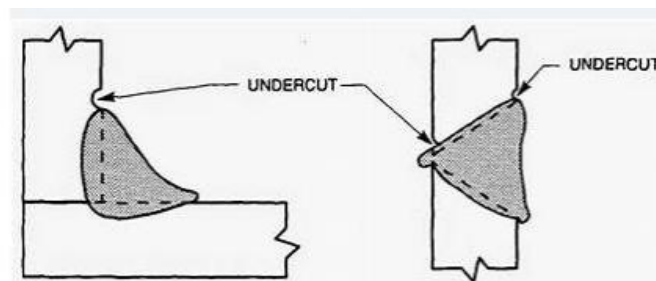
2. Lack of Fusion: Incomplete joining of base metal.



3. Slag Inclusion: Trapped slag particles.



4. Undercut: Groove or depression at weld toe.



ACTIVITY 15
KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Welding Terminology and International Standards:

1. What is the definition of welding according to AWS?

- A) Joining two materials by heat
- B) Fabrication process joining materials by heat, pressure, or both
- C) Melting metal to form a joint
- D) Cutting metal with a torch

2. What international standard governs welding terminology?

- A) ISO
- B) AWS
- C) ASME
- D) All of the above

3. What is the term for a tubular wire electrode with flux core?

- A) Solid wire
- B) Cored-wire
- C) Flux-cored wire
- D) Metal-cored wire

4. What welding process uses a cored-wire electrode?

- A) GMAW
- B) GTAW
- C) SMAW
- D) FCAW

5. What is the purpose of shielding gas in welding?

- A) To generate heat
- B) To protect the arc and weld area
- C) To regulate wire feed speed
- D) To cool the wire electrode

6. What is the term for cavities or pores in a weld?
- A) Lack of fusion
 - B) Porosity
 - C) Slag inclusion
 - D) Undercut
7. What international standard governs welding safety?
- A) OSHA
 - B) ASME
 - C) ISO
 - D) AWS
8. What is the term for incomplete joining of base metal?
- A) Lack of fusion
 - B) Porosity
 - C) Slag inclusion
 - D) Undercut
9. What is the purpose of ventilation in welding?
- A) To generate shielding gas
 - B) To regulate wire feed speed
 - C) To remove fumes and particles
 - D) To maintain gas pressure
10. What organization publishes standards for welding in the United States?
- A) AWS
 - B) ASME
 - C) ISO
 - D) OSHA

ASSESSMENT:

1. Learners identify and describe welding equipment components.
2. Learners explain function of each component.
3. Learners demonstrate equipment setup and operation.

Welding Consumables Storage:

1. Electrodes: Store in dry, protected area to prevent moisture damage.



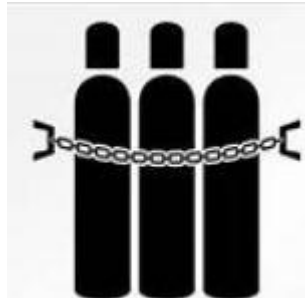
2. Wire feed rolls: Store in designated wire rack or cabinet.



3. Contact tips: Store in small parts container or designated tip holder.



4. Shielding gas cylinders: Secure upright, chained, and labeled.



TOOLS AND EQUIPMENT STORAGE:

1. Welding machines: Store in designated area, covered, and secured.
2. Welding guns: Store in designated gun rack or holder.
3. Welding cables: Store on cable reels or designated cable hooks.
4. PPE (helmets, gloves, etc.): Store in designated PPE storage area.
5. Hand tools (pliers, wrenches, etc.): Store in designated toolboxes.

Activity 16

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Storage of Welding Consumables and Equipment:

1. Where should welding electrodes be stored?

- A) Humid area
- B) Dry, protected area
- C) Outdoor storage
- D) Near welding machine

2. What is the purpose of labeling storage areas?

- A) To prevent theft
- B) To improve productivity
- C) To ensure organization
- D) To reduce costs

3. How should shielding gas cylinders be stored?

- A) Lying down
- B) Upright, chained, and labeled
- C) Near welding machine
- D) In humid area

4. What is the benefit of storing wire feed rolls in a designated wire rack?

- A) Reduced costs
- B) Improved productivity
- C) Prevents damage
- D) All of the above

5. Where should welding guns be stored?

- A) Near welding machine
- B) Designated gun rack or holder
- C) Outdoor storage
- D) Humid area

6. Why is it important to secure equipment and consumables?

- A) To prevent theft
- B) To prevent damage
- C) To improve productivity
- D) All of the above

7. What is the purpose of storing hand tools in designated toolboxes?

- A) To reduce costs
- B) To improve productivity
- C) To prevent loss
- D) All of the above

8. How should welding cables be stored?

- A) On floor
- B) On cable reels or designated cable hooks
- C) Near welding machine
- D) In humid area

9. What is the benefit of storing PPE in a designated area?

- A) Reduced costs
- B) Improved productivity
- C) Prevents damage
- D) Easy access

10. Why is proper storage important in a welding worksite?

- A) To reduce costs
- B) To improve productivity
- C) To ensure safety
- D) All of the above

**STEP-BY-STEP GUIDE ON PREPARING WORK-PIECES PRIOR TO WELDING AND
WELDING THEM:**

PREPARING WORK-PIECES:

1. Cleanliness: Remove dirt, oil, grease, and other contaminants.
2. Deburring: Remove sharp edges and burrs.
3. Surface preparation: Grind or sand to achieve desired surface finish.

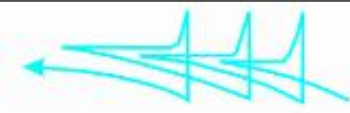


4. Alignment: Ensure accurate alignment of work-pieces.
5. Clamping: Secure work-pieces in place.
6. Measuring: Verify dimensions and tolerances.

Welding Work-pieces:

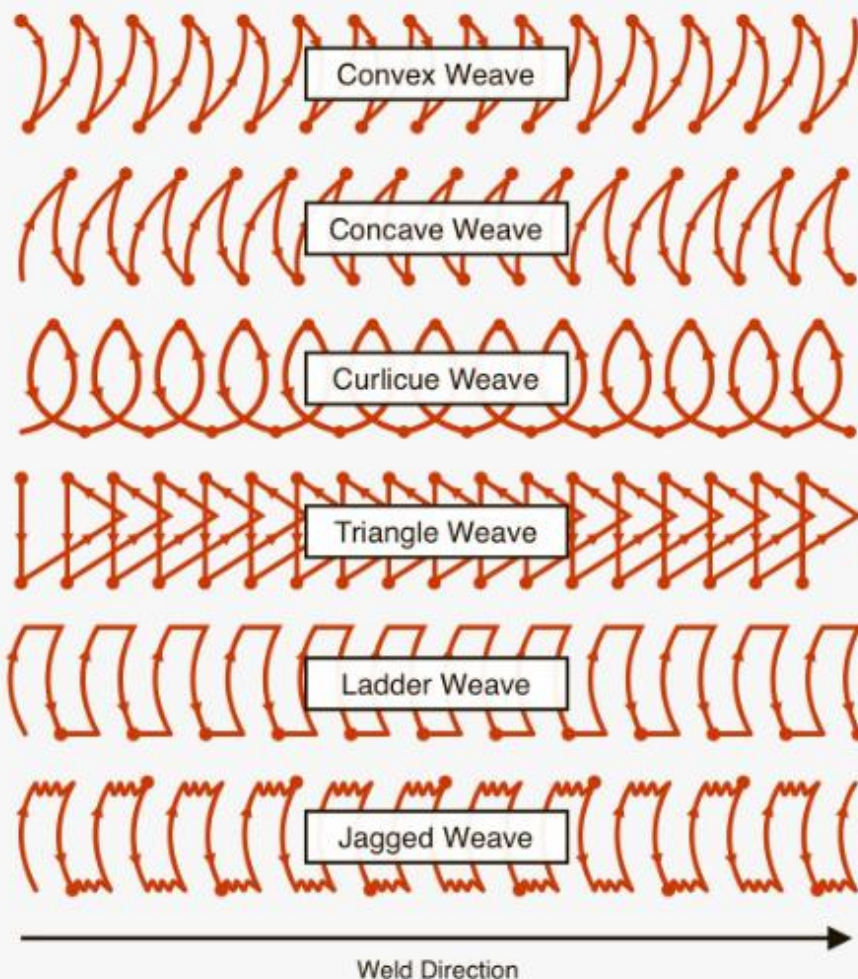


1. Select welding process (GMAW, GTAW, FCAW, etc.).
2. Choose welding parameters (voltage, current, wire feed speed).
3. Set up welding equipment (power source, wire feeder, gun).
4. Don personal protective equipment (PPE).
5. Start welding, maintaining consistent arc length and speed.
6. Monitor weld pool and adjust parameters as needed.

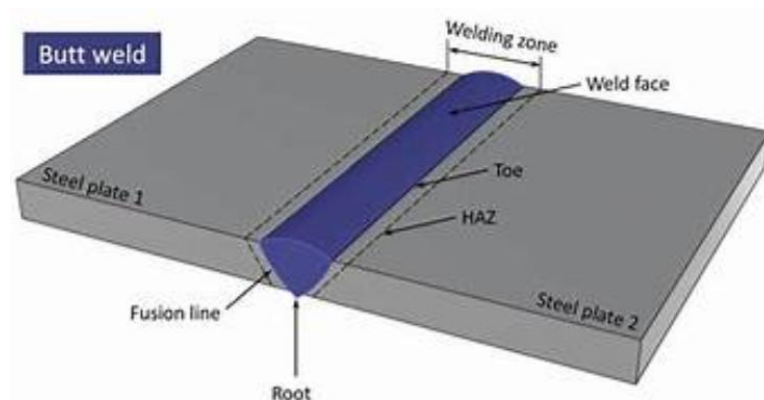
Welding Techniques:	
 C	 J
 Figure 8	 T
 V	 Square

 Circular	 Upside Down V
 Straight Stepped	 Zigzag

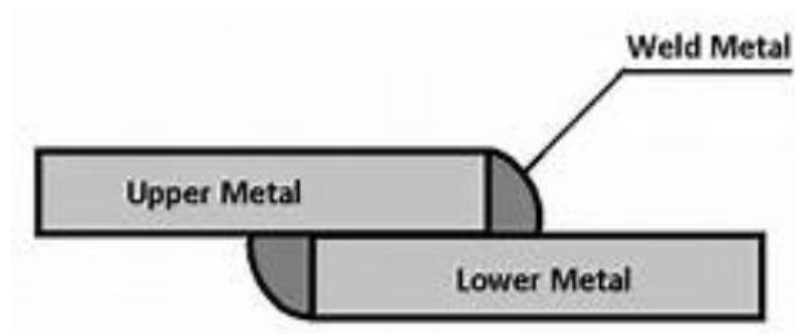
Welding Weaving Patterns



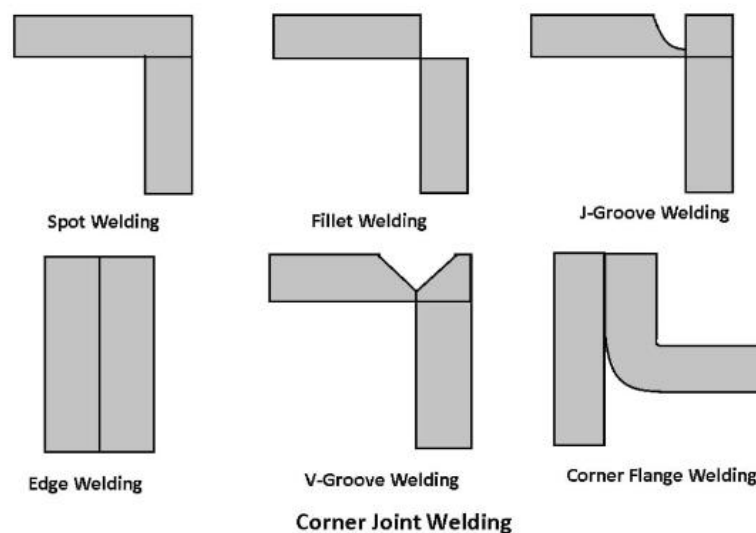
1. Butt joint: Weld two pieces end-to-end.



2. Lap joint: Weld overlapping pieces.



3. Corner joint: Weld pieces at 90-degree angle.



ACTIVITY 17

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

1. What is the first step in preparing work-pieces for welding?

- A) Deburring
- B) Cleaning
- C) Measuring
- D) Clamping

2. What welding process is commonly used for thin materials?

- A) GMAW
- B) GTAW
- C) FCAW
- D) SMAW

3. What is the purpose of welding parameters?

- A) To select welding process
- B) To set up welding equipment
- C) To control weld quality
- D) To ensure safety

4. What is the purpose of clamping work-pieces?

- A) To ensure accuracy
- B) To prevent movement
- C) To maintain alignment
- D) All of the above

5. What type of joint is used to weld two pieces end-to-end?

- A) Butt joint
- B) Lap joint
- C) Corner joint
- D) Edge joint

CHAPTER 4

SECTOR IDENTIFICATION AND ORGANIZATION TYPE IN SOUTH AFRICA:

A sector is a specific area of economic or industrial activity. Common sectors include:

Sector/occupational focus: Services, manufacturing, financial, educational etc

Sectors in South Africa:

1. Agriculture, Forestry, and Fishing
2. Mining and Quarrying
3. Manufacturing
4. Electricity, Gas, and Water Supply
5. Construction
6. Wholesale and Retail Trade
7. Transportation, Storage, and Communication
8. Financial Intermediation, Real Estate, and Business Services
9. Community, Social, and Personal Services
10. Tourism

Organization type refers to the structure, ownership, and goals of an entity:

1. Private Sector: Businesses, corporations, startups, etc.
2. Public Sector: Government departments, state-owned enterprises (SOEs), etc.
3. Non-Profit Organizations (NPOs): Charities, foundations, associations, etc.
4. Small, Medium, and Micro Enterprises (SMMEs): Businesses with fewer than 500 employees.
5. Cooperatives: Member-owned businesses.

Industry Classification Systems in South Africa:

1. Standard Industrial Classification (SIC)
2. South African National Treasury's Economic Classification System
3. Statistics South Africa's (Stats SA) Industrial Classification System

Key Sectors in South Africa:

1. Financial Services: Banking, insurance, investments
2. Manufacturing: Automotive, textiles, food processing
3. Mining: Gold, platinum, coal, iron ore
4. Tourism: Hospitality, travel, tourism services
5. Agriculture: Crop production, livestock farming

Major Organizations in South Africa:

1. Private Sector: Shoprite, MTN, Sasol
2. Public Sector: Eskom, Transnet, South African Airways

- 3. Non-Profit: Nelson Mandela Foundation, Red Cross South Africa
- 4. SMMEs: Small businesses in various sectors

ACTIVITY 18

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Multiple-Choice Questions:

1. Which sector is the largest contributor to South Africa's GDP?
 - A) Manufacturing
 - B) Financial Services
 - C) Mining
 - D) Agriculture
2. What type of organization is the Nelson Mandela Foundation?
 - A) Private Sector
 - B) Public Sector
 - C) Non-Profit
 - D) SMME
3. Which classification system is used by Stats SA?
 - A) SIC
 - B) South African National Treasury's Economic Classification System
 - C) Stats SA's Industrial Classification System
 - D) ISIC
4. Which sector is responsible for the production of goods in South Africa?
 - A) Services
 - B) Manufacturing
 - C) Financial
 - D) Agriculture
5. What type of organization is Eskom?
 - A) Private Sector
 - B) Public Sector
 - C) Non-Profit

D) SMME

6. Which industry classification system is used by the South African National Treasury?

- A) SIC
- B) ISIC
- C) South African National Treasury's Economic Classification System
- D) Stats SA's Industrial Classification System

7. Which sector is the largest employer in South Africa?

- A) Manufacturing
- B) Financial Services
- C) Government
- D) Tourism

7. What is the primary function of the Companies and Intellectual Property Commission (CIPC)?

- A) Regulate financial markets
- B) Register businesses
- C) Enforce labor laws
- D) Promote tourism

8. Which organization is responsible for setting standards in South Africa?

- A) South African Bureau of Standards (SABS)
- B) National Treasury
- C) Department of Trade, Industry, and Competition
- D) South African Reserve Bank

9. Which sector contributes significantly to South Africa's export revenue?

- A) Manufacturing
- B) Mining
- C) Agriculture
- D) Tourism

10. What type of organization is a cooperative?

- A) Private Sector
- B) Public Sector
- C) Non-Profit
- D) Member-owned business

11. Which regulatory body oversees the banking sector in South Africa?

- A) South African Reserve Bank
- B) National Treasury
- C) Department of Trade, Industry, and Competition
- D) Financial Sector Conduct Authority

12. Which sector is crucial for South Africa's economic growth and development?

- A) Financial Services
- B) Manufacturing
- C) Tourism
- D) All of the above

PROJECTED INCOME FOR WELDING PROJECTS

1. Department of Basic Education does the funding
2. Private sector sponsorships
3. Parent-Teacher Association (PTA) contributions
4. Sales of welded products (e.g., furniture, decorative items)
5. Training and certification fees (e.g., for learnerships)

Projected Expenditure:

1. Equipment and machinery (e.g., welding machines, consumables)
2. Materials (e.g., steel, aluminum, welding rods)
3. Safety gear (e.g., helmets, gloves, goggles)
4. Instructor salaries and training
5. Facility rental or maintenance
6. Marketing and advertising
7. Insurance (liability, equipment)

REALISTIC PROJECTION EXAMPLE (ANNUAL):			
PROJECTED INCOME:			
1)	Government funding:	R200,000	

2)	Private sponsorships:	R50,000	
3)	PTA contributions:	R20,000	
4)	Sales of welded products:	R30,000	
5)	Training fees:	R10,000	
6)	Total income:	R310,000	
Projected Expenditure			
7)	Equipment and machinery:		R100,000
8)	Materials:		R50,000
9)	Safety gear:		R10,000
10)	Instructor salaries:		R120,000
11)	Facility rental:		R20,000
12)	Marketing:		R10,000
13)	Insurance:		R5,000
14)	Total expenditure:		R315,000
Net Income (Annual):		R310,000-R315,000 = -R5,000 (loss)	

R310,000 (income) - R315,000 (expenditure) = -R5,000

Key Considerations:

1. Fluctuating government funding
2. Competition for private sponsorships
3. Limited parental contributions
4. High equipment and material costs
5. Safety and liability concerns
6. Instructor expertise and training requirements
7. Marketing and advertising challenges

1. Excel spreadsheets
2. Financial planning software (e.g., QuickBooks, Xero)

ACTIVITY 19

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Projected income and expenditure for welding projects in South African schools:

Question 1

What is a potential source of income for a school welding project?

- A) Government funding
- B) Private sponsorships
- C) Sales of welded products
- D) All of the above

Question 2

What is a significant expenditure for a school welding project?

- A) Instructor salaries
- B) Marketing and advertising
- C) Equipment and machinery
- D) Insurance

Question 3

Which organization provides guidelines for school workshops in South Africa?

- A) Department of Higher Education
- B) Department of Basic Education
- C) South African Qualifications Authority
- D) National Treasury

Question 4

What is a key consideration when creating projected income and expenditure for a school welding project?

- A) Fluctuating government funding
- B) High equipment costs
- C) Safety concerns
- D) All of the above

Question 5

What is the purpose of creating projected income and expenditure for a school welding project?

- A) To secure government funding
- B) To determine profitability
- C) To plan and manage financial resources
- D) To attract private sponsorships

DEMONSTRATE AND DEFINE INITIAL COST FOR A METAL BRAAI STAND BEFORE WORK HAS STARTED AND PROFIT THAT CAN BE ACHIEVED WHEN A PRODUCT IS COMPLETED.

The initial cost and potential profit for a metal braai stand:

Initial Cost (Before Work Starts)			
Materials	2mm steel plate (1.5m x 1.5m)	R500	
	Steel tubing (10mm x 10mm x 6m)	R300	
	Welding rods:	R100	
	Paint and coating:	R200	
	Total materials cost		R1,100
Labor	Welder's hourly rate:	R150/hour	
	Estimated labor hours:	10 hours	
	Total labor cost:		R1,500
Overheads	Workshop rental:	R500	
	Electricity	R100	
	Total overheads:		R600
	Total Initial Cost:		R3,200

Selling Price (After Completion)

Profit Calculation			
	Estimated selling price: R6,000		
	Selling Price	R6,000	
	Total Initial Cost	-R3,200	
Profit			R2,800

Profit Margin:

Profit Margin = (Profit / Selling Price) x 100

Profit Margin = (R2,800 / R6,000) x 100

Profit Margin = 46.67%

Break-Even Analysis:

Break-Even Point = Total Initial Cost / (Selling Price - Variable Costs)

Break-Even Point = R3,200 / (R6,000 - R1,100)

Break-Even Point = 6 units

This means that if you sell 6 units of the metal braai stand, you will break even and start making a profit.

Assumptions:

- The selling price remains constant.
- Material and labor costs remain constant.
- No additional expenses or losses.

Note: These figures are estimates and may vary depending on location, market conditions, and other factors.

ACTIVITY 20

KWA-THEMA SKILLS SCHOOL

Learner Name:..... Surname:

Term : 4 Date :

Initial cost and profit for a metal braai stand:

Question 1

What is included in the initial cost of producing a metal braai stand?

- A) Materials, labor, and overheads
- B) Materials and labor only
- C) Overheads and marketing expenses
- D) Selling price and profit

Question 2

What is the estimated selling price of a metal braai stand?

- A) R3,000
- B) R6,000
- C) R9,000
- D) R12,000

Question 3

What is the profit margin if the selling price is R6,000 and the initial cost is R3,200?

- A) 20%
- B) 46.67%
- C) 60%

D) 80%

Question 4

What is the break-even point if the initial cost is R3,200 and the selling price is R6,000?

- A) 4 units
- B) 6 units
- C) 8 units
- D) 10 units

Question 5

Which of the following affects the profit margin of a metal braai stand?

- A) Material costs only
- B) Labor costs only
- C) Selling price and material costs
- D) All of the above

Question 6

What is the purpose of calculating the break-even point for a metal braai stand?

- A) To determine profitability
- B) To set selling price
- C) To calculate initial cost
- D) To identify production volume for break-even

Question 7

Which expense is considered a variable cost in producing a metal braai stand?

- A) Rent
- B) Electricity
- C) Materials
- D) Labor

Question 8

What happens to profit margin if selling price increases?

- A) Decreases
- B) Remains constant
- C) Increases
- D) Unaffected

Question 9

Which factor affects initial cost of producing a metal braai stand?

- A) Market demand

- B) Competition
- C) Material availability
- D) All of the above

Question 10

What is the benefit of calculating profit margin for a metal braai stand?

- A) Identifies areas for cost reduction
- B) Helps set selling price
- C) Evaluates business performance
- D) All of the above

CHAPTER 5

REVISION OF WORK COVERED DURING THE COURSE

Revise and reflecting, together with all relevant and related materials, wall charts, notes and power point presentations

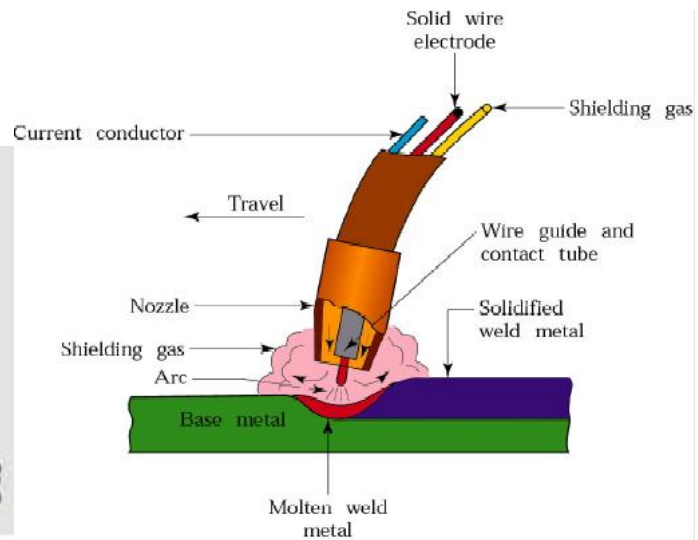
Safety and First Aid		
1. Personal Protective Equipment (PPE) - Safety glasses - Welding helmet - Gloves - Steel-toed boots	2. Hazards in Welding - Electric shock - Burns - Fire - Inhalation of fumes	3. First Aid Procedures - Burns - Cuts - Electric shock - Fire extinguishers

TOOLS		Materials and Joining	
1. Hand Tools - Pliers - Wrenches - Hammers - Tape measures	2. Power Tools - Grinders - Drills - Sanders - Cutting machines	1. Metals - Steel - Aluminum - Copper - Brass	2. Joining Techniques - Welding - Brazing - Soldering - Riveting

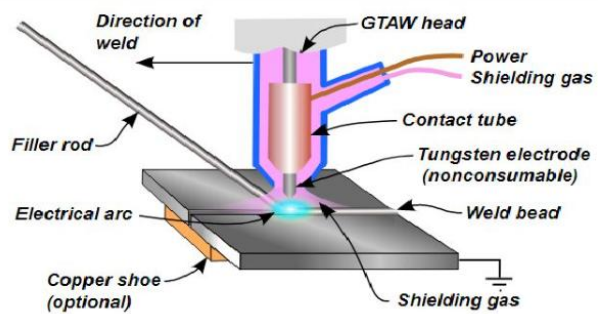
Welding Equipment

1. Welding Machines

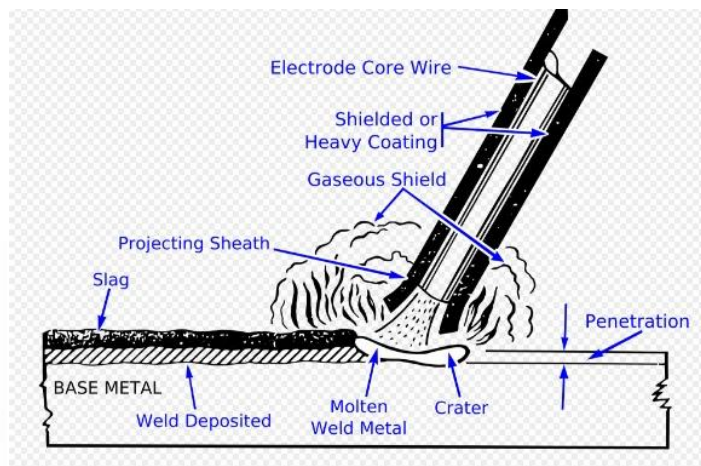
MIG (GMAW)



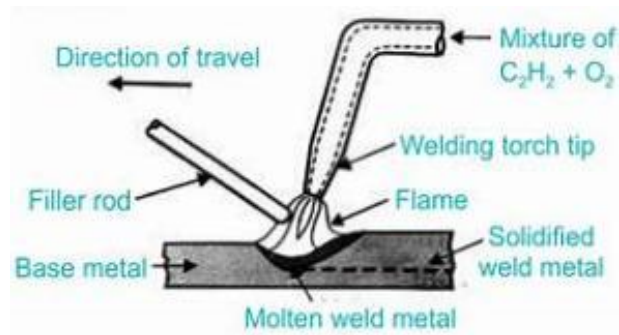
TIG (GTAW)



Arc (SMAW)



Gas Welding (004Fy-acetylene)



2. Consumables

- Electrodes
- Filler metals
- Shielding gases

3. Welding Processes

REVISION MATERIALS

WORKSHOP PRACTICES

MATERIAL SELECTIONS

1. Metal Properties

- Ferrous and non-ferrous metals
- Strength, durability, and corrosion resistance

2. Material Classification

- Steel, aluminum, copper, and brass
- Wood, plastic, and composite materials

3. Material Selection Criteria

- Cost, availability, and sustainability
- Strength, weight, and aesthetic considerations

Drawings and Shapes

1. Technical Drawing

- Orthographic projections
- Isometric drawings
- Dimensioning and tolerancing

2. Geometric Shapes

- Points, lines, and planes
- Angles, curves, and surfaces
- 3. Blueprint Reading
 - Symbols, abbreviations, and notation
 - Scale and proportion

ASSESSMENT STRUCTURE

1. Practical Task(s) (75% weighting)
 - Assess practical skills and competence in welding processes
 - Task(s) will be completed during scheduled practical sessions
 - Assessment criteria will include:
 - Quality of workmanship
 - Safety procedures
 - Efficiency and productivity
 - Technical accuracy
1. Theory Test (25% weighting)
 - Assess knowledge and understanding of welding principles and theory
 - Test will consist of multiple-choice questions and/or short-answer questions
 - Assessment criteria will include:
 - Understanding of welding processes and techniques
 - Knowledge of safety procedures and protocols
 - Familiarity with welding equipment and tools
 - Ability to apply theoretical knowledge to practical situations

Assessment Schedule

- Week 5: Practical Task 1 (25% of practical mark) Group work
- Week 10: Practical Task 2 (25% of practical mark) individual with teachers' help
- Week 15: Practical Task 3 (25% of practical mark) timed PAT
- Week 18: Theory Test (25% of total mark) End of term test

Assessment Criteria

Practical Tasks:

- Quality of workmanship (30%)
- Safety procedures (20%)
- Efficiency and productivity (20%)
- Technical accuracy (30%)

Theory Test

- Understanding of welding processes and techniques (40%)
- Knowledge of safety procedures and protocols (30%)

- Familiarity with welding equipment and tools (20%)
- Ability to apply theoretical knowledge to practical situations (10%)

Grading

- Distinction: 80-100%
- Merit: 60-79%
- Pass: 50-59%
- Fail: below 50%

Assessment Rubric:

Criteria	Excellent (90-100%)	Good (80-89%)	Satisfactory (70-79%)	Needs Improvement (Below 70%)
Measuring	Accurate, precise	Mostly accurate	Some errors	Inaccurate
Cutting	Smooth, safe	Some imperfections	Rough, unsafe	Hazardous
Welding	Strong, penetration	Good, minor flaws	Weak, porous	Poor, incomplete
Grinding	Smooth, even	Some scratches	Uneven, rough	Poor finish
Drilling	Accurate, precise	Some deviation	Inaccurate	Off-center
Safety	Excellent PPE, tool handling	Good, minor lapses	Some safety concerns	Hazardous behavior
Quality	Excellent finish, presentation	Good, minor flaws	Satisfactory, some imperfections	Poor, incomplete

Grading:

- Distinction: 90-100%
- Merit: 80-89%
- Pass: 70-79%
- Fail: Below 70%

Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed.

- Measuring
- Cutting
- Welding
- Drilling
- Grinding

Total for Activity 1 Marks to be converted to 25% of the total term mark.

PRACTICAL ASSESSMENT TASK

Activity 1

Measuring, Cutting, Welding, Drilling, and Grinding

Total Marks: 100

Weighting: 25% of Total Term Mark

LEARNER NAME:		DATE:				
Measuring (15 marks)		MARKS				
Assessment Criteria		1	2	3	4	5
Accuracy (5 marks)	✧ Measure two pieces of material to required dimensions ✧ Tolerance: $\pm 1\text{mm}$					
Precision (5 marks)	✧ Measure and mark reference points for cutting and drilling ✧ Clarity and legibility of markings					
Use of measuring tools (5 marks)	Correct use of measuring instruments (e.g., tape measure, calipers)					

LEARNER NAME:		DATE:				
Cutting (15 marks)		MARKS				
Assessment Criteria		1	2	3	4	5
Technique (5 marks)	Clean, straight cut Minimal burrs or rough edges					
Safety (5 marks)	Proper use of cutting tools and PPE Safe handling of materials					
Quality of cut (5 marks)	Smooth, even surface finish					

Welding (25 marks)

LEARNER NAME:		DATE:									
Welding (25 marks)		MARKS									
Assessment Criteria		1	2	3	4	5	6	7	8	9	10
Tack welds (10 marks)	Strong, mechanically sound Proper penetration and fusion										
Fillet weld (10 marks)	Minimum 30mm length Consistent, even bead Proper penetration and fusion										
Weld appearance (5)	Smooth, even surface finish										

